

A counterexample to Conjecture 00000002304

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The source proposes an equality classification under *general linear actions*: equality in $k(GV) \leq k(G)k(V)$ occurs exactly when, among other conditions, the G -module V is faithful. Here $k(H)$ denotes the number of conjugacy classes of a finite group H , and $k(V)$ refers to the additive group of the vector space. We give a general linear action attaining equality that is not faithful. Both language versions state the general-action scope; neither imposes faithfulness or coprimality as a hypothesis.

Construction. Let $G = C_2 = \{0, 1\}$ with addition modulo 2 and identity 0. Let the field be $\mathbb{F}_2 = \{0, 1\}$, with its usual addition and multiplication modulo 2, and put $V = \mathbb{F}_2$, viewed as a one-dimensional vector space over itself. Define the action by

$$\rho(g)(v) = v \quad (g \in G, v \in V).$$

Every action map is the identity linear automorphism; the identity and composition laws for a group action hold. Thus this is a legitimate linear action. Its kernel is all of the nontrivial group G , so it is not faithful. Equivalently, $\rho(0) = \rho(1)$ while $0 \neq 1$. The vector space is nonzero and one-dimensional; in particular, the failure of faithfulness is not an artifact of using the zero module.

Use the standard semidirect product on the carrier $V \times G$:

$$(v, g)(w, h) = (v + \rho(g)(w), g + h) = (v + w, g + h).$$

Consequently $V \rtimes_\rho G = C_2 \times C_2$, the four-element abelian group. Its identity is $(0, 0)$ and each element is its own inverse.

Conjugacy classes and contradiction. In an abelian group, conjugating x by any g gives $gxg^{-1} = x$. Hence each element is a singleton conjugacy class. Therefore

$$k(G) = 2, \quad k(V) = 2, \quad k(V \rtimes_\rho G) = 4 = 2 \cdot 2 = k(G)k(V).$$

Equality holds, yet V is not a faithful G -module. This contradicts the faithfulness condition required by the proposed classification. It also contradicts the claim of strict inequality for all remaining actions. No interpretation of the additional character-degree assertion can restore this false necessary condition.

Scope. This disproof addresses the conjecture exactly as written, which quantifies over general linear actions. One could ask a different question after imposing a faithful-action hypothesis, or further hypotheses such as coprimality. Our example is deliberately not faithful, and $|G| = 2$ equals the characteristic of the field, so it is not offered as a counterexample to either of those restricted variants. The source's stated equality classification cannot itself supply a missing assumption excluding counterexamples to that classification.

Formal proof and correspondence with the mathematical objects. The file `lean/Main.lean` is self-contained over `Std`, using Lean 4.19.0.

`GroupData` contains multiplication, identity, inverse, and all group laws. `FieldData` contains the additive and multiplicative field laws, distributivity, inverses of all nonzero elements, and the axiom $0 \neq 1$. `VectorSpaceData` contains an abelian additive group and the vector-space scalar laws. The concrete values `C2`, `F2`, and `V2` verify these axioms with the modulo-2 operations on `Fin 2`. The results `every_vector_is_scalar` and `basis_vector_nonzero` additionally exhibit the nonzero spanning vector `1`.

`LinearAction` asserts the group action laws and preservation of addition and scalar multiplication. The theorem `action_inverse` records the inverse-action identity, so the acting maps are indeed automorphisms. `trivialAction` is the action used above. `Faithful` is injectivity of the map from group elements to action maps, and `not_faithful` uses the distinct elements `0` and `1` directly.

The multiplication of the group `GV`, defined on the entire product `Fin 2 × Fin 2`, is precisely the displayed semidirect formula. All group laws are verified; the theorems `semidirect_multiplication` and `semidirect_identity` expose that correspondence.

In particular, the formalization does not replace conjugacy-class counts by unrelated constants. `Conjugate H x y` is defined by the existence of a group element g with $gxg^{-1} = y$. `HasClassNumber H n` requires a setoid with exactly this relation and a bijection from `Fin n` onto its actual `Quotient`. The formal proof verifies that conjugacy is equality in each of the three groups. It then builds explicit quotient bijections, including a complete four-element enumeration of the product carrier, proving the class numbers `2, 2, 4`.

`ClaimedEqualityNecessity` is the faithfulness implication required by the source, restricted to the specific finite group, field, and vector space already constructed, but quantified over their linear actions and corresponding semidirect products. A universal general-action equality classification necessarily implies this restricted claim. `counterexample` states the class-number equalities and failure of faithfulness together; `conjecture2304_false` proves the negation of that necessary implication.

The proof uses kernel-reduced `decide`, with no `native_decide`, admitted statement, or custom axiom. Printed axioms of the final theorem are exactly `propext`, `Classical.choice`, and `Quot.sound`.